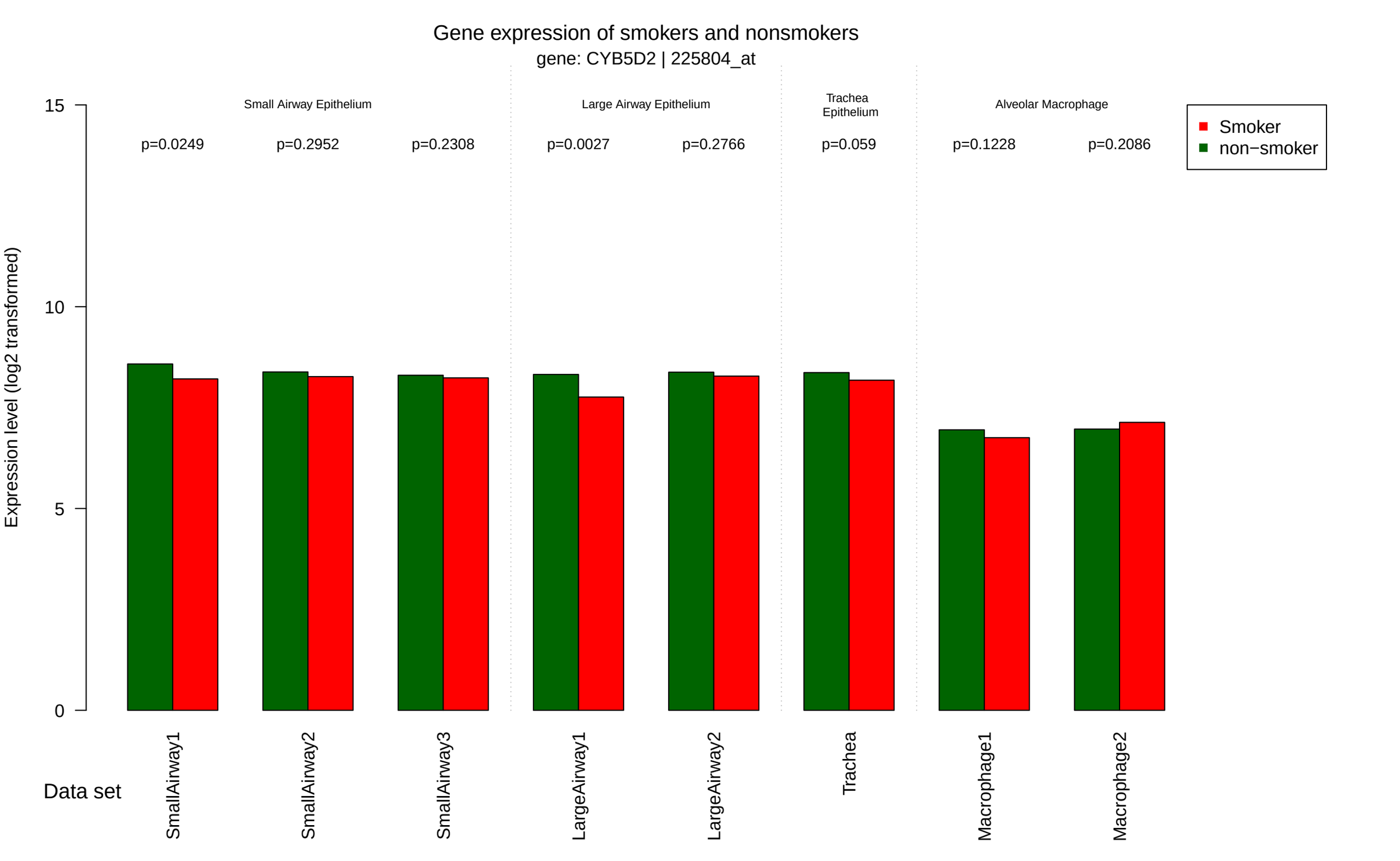
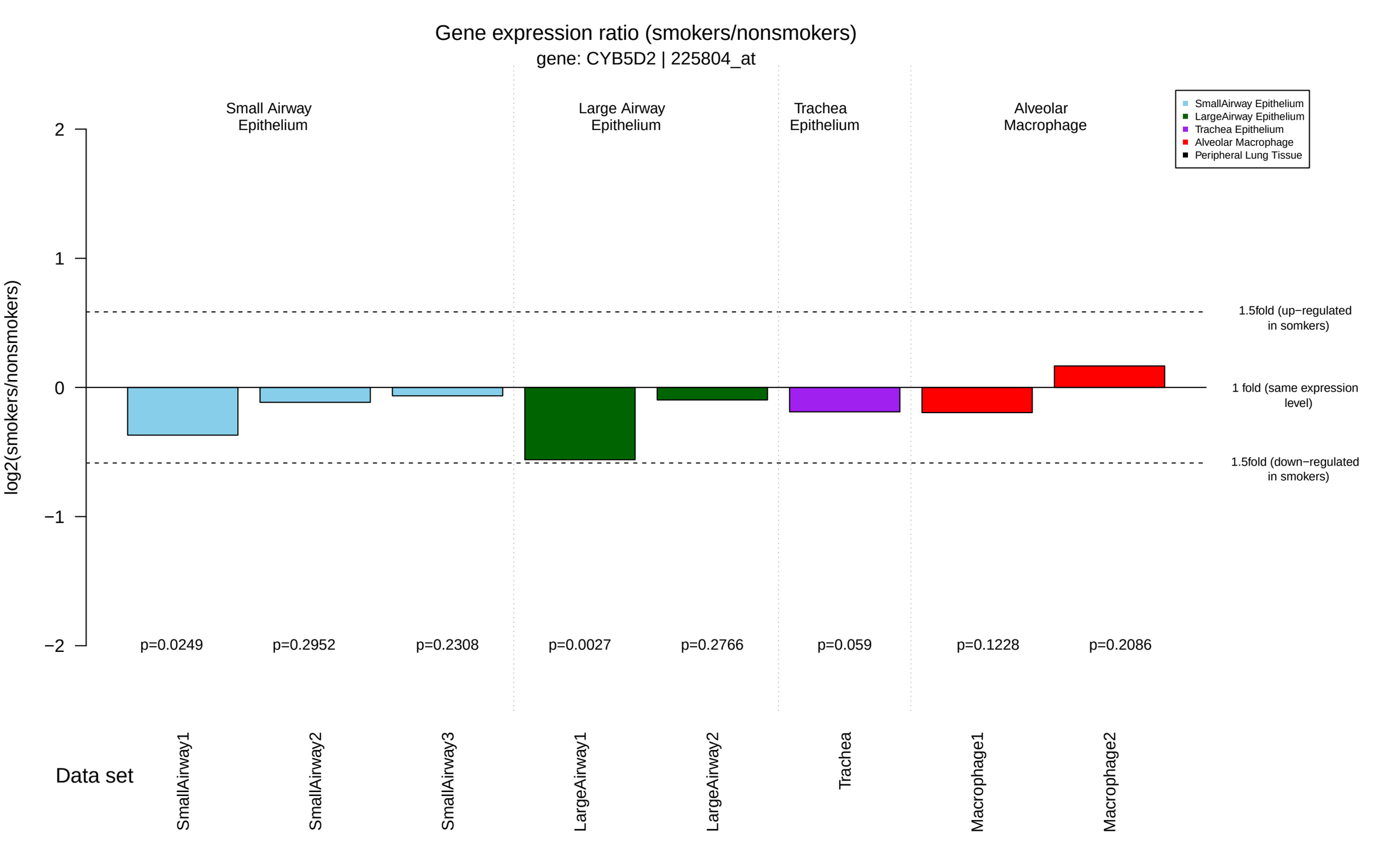
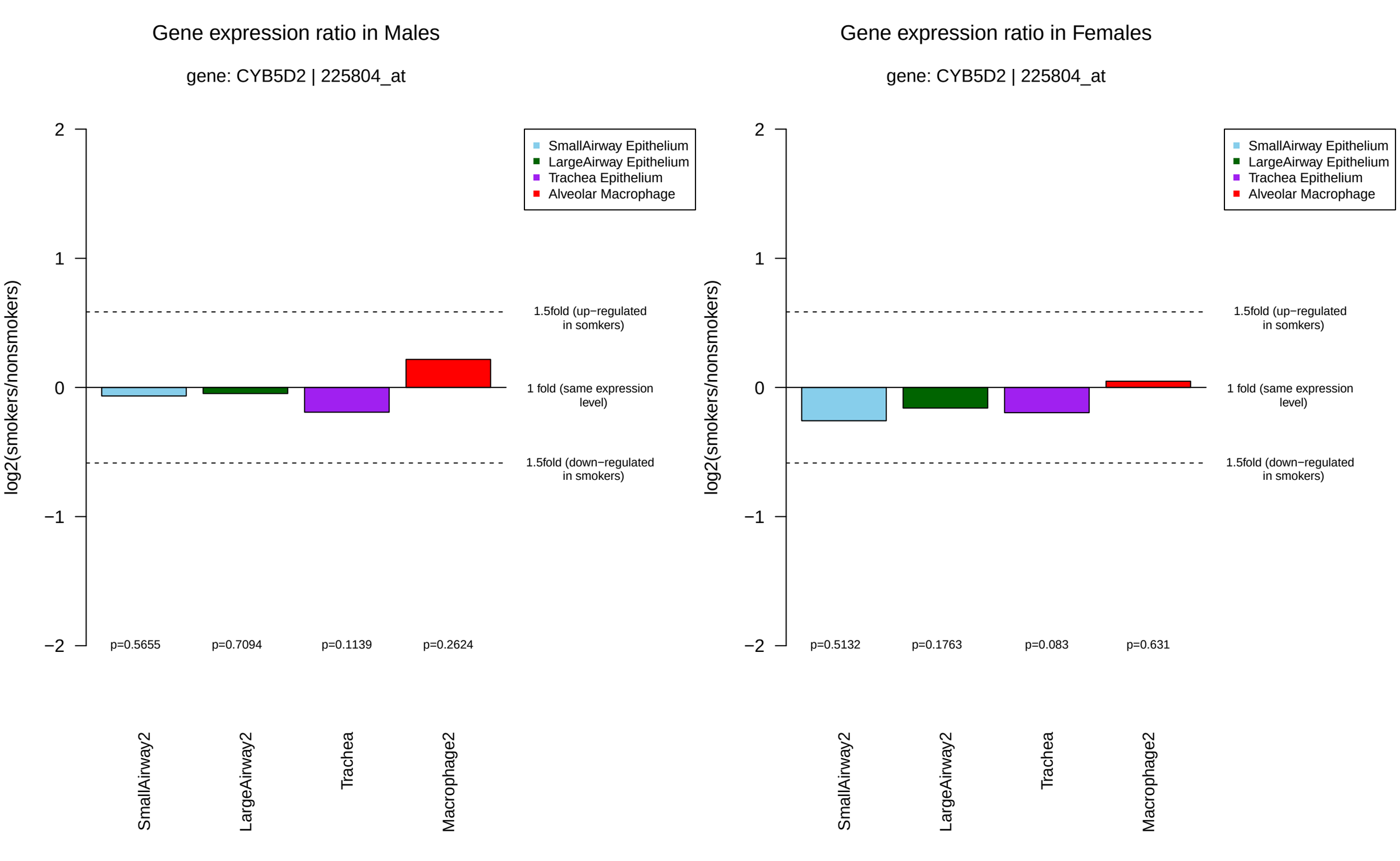
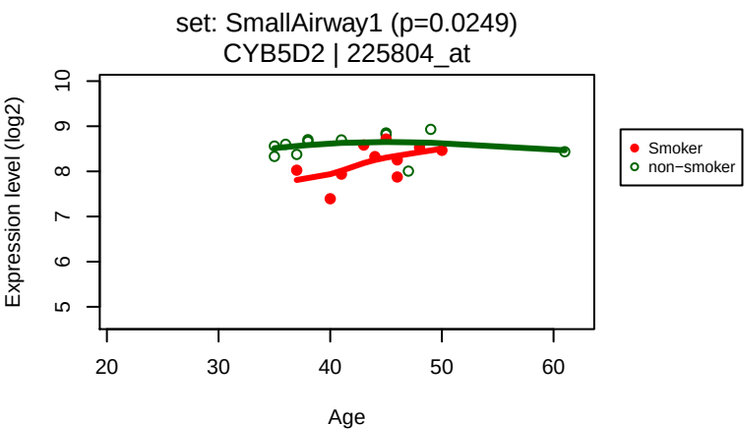
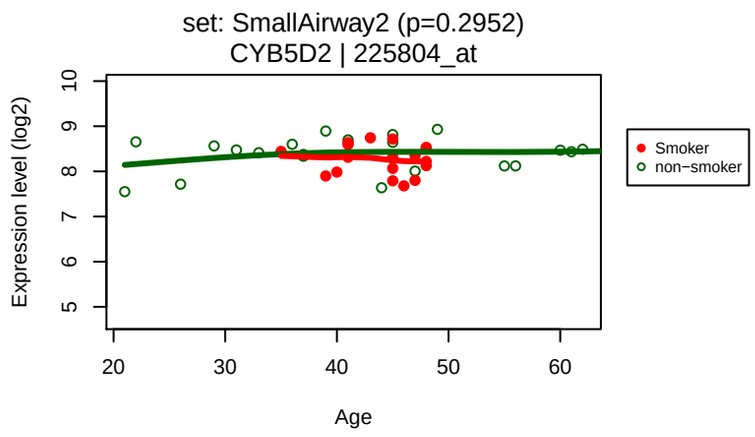
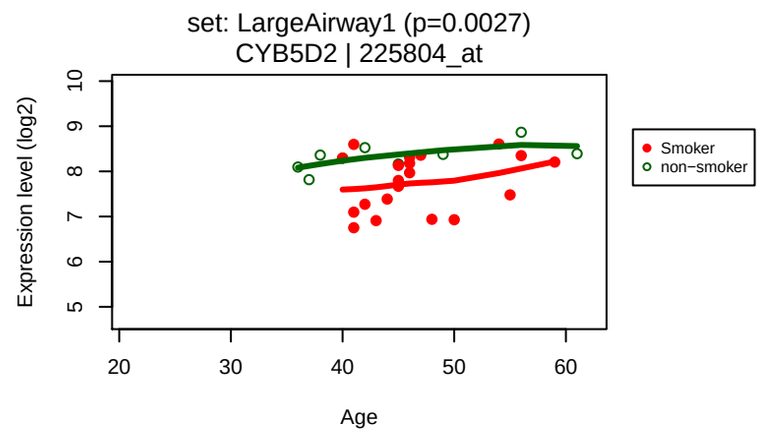

Gene expression of smokers and nonsmokers
gene: CYB5D2 | 225804_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0249
p=0.2952
p=0.2308
p=0.0027
p=0.2766
p=0.059
p=0.1228
p=0.2086
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

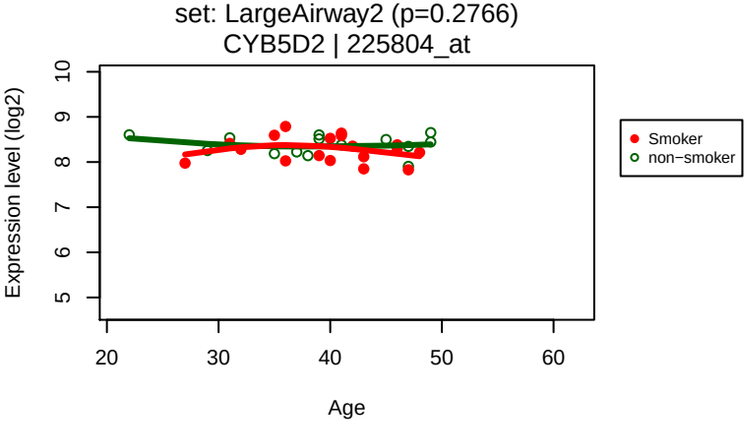

Gene expression ratio (smokers/nonsmokers)
gene: CYB5D2 | 225804_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0249
p=0.2952
p=0.2308
p=0.0027
p=0.2766
p=0.059
p=0.1228
p=0.2086
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

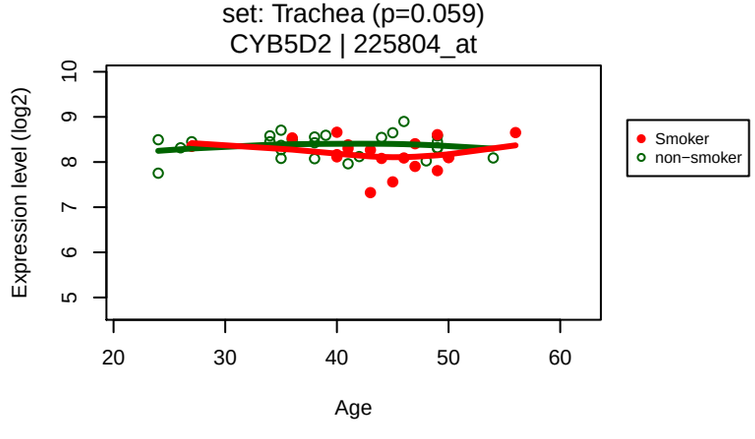

Gene expression ratio in Males
gene: CYB5D2 | 225804_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5655
p=0.7094
p=0.1139
p=0.2624
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYB5D2 | 225804_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5132
p=0.1763
p=0.083
p=0.631
SmallAirway2
LargeAirway2
Trachea
Macrophage2

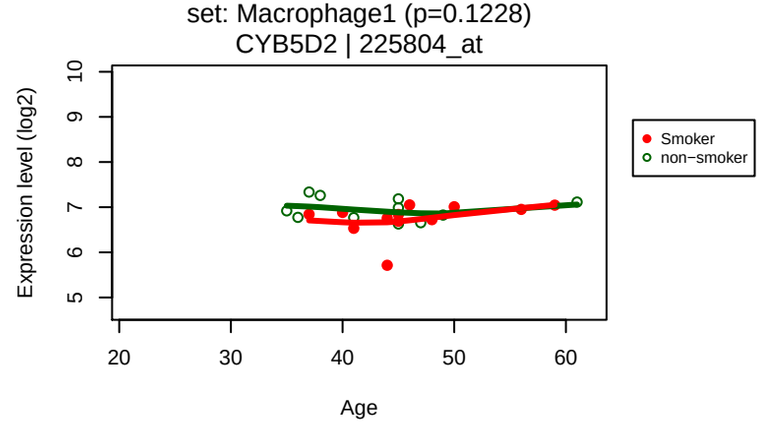

set: SmallAirway1 (p=0.0249)
CYB5D2 | 225804_at
Expression level (log2)
Age
Smoker
non−smoker

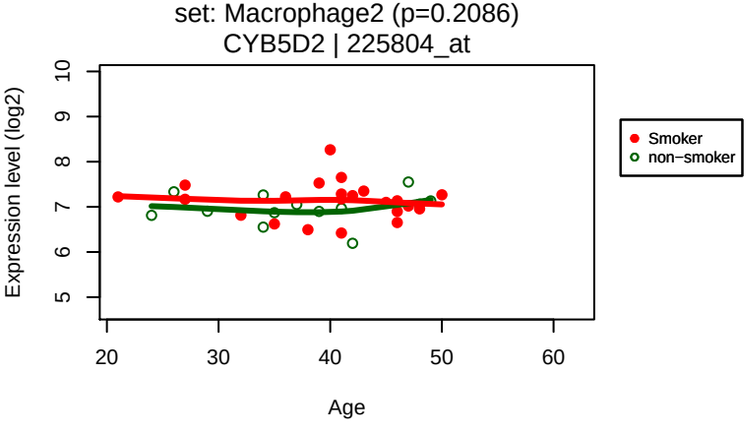

set: SmallAirway2 (p=0.2952)
CYB5D2 | 225804_at
Expression level (log2)
Age
Smoker
non−smoker

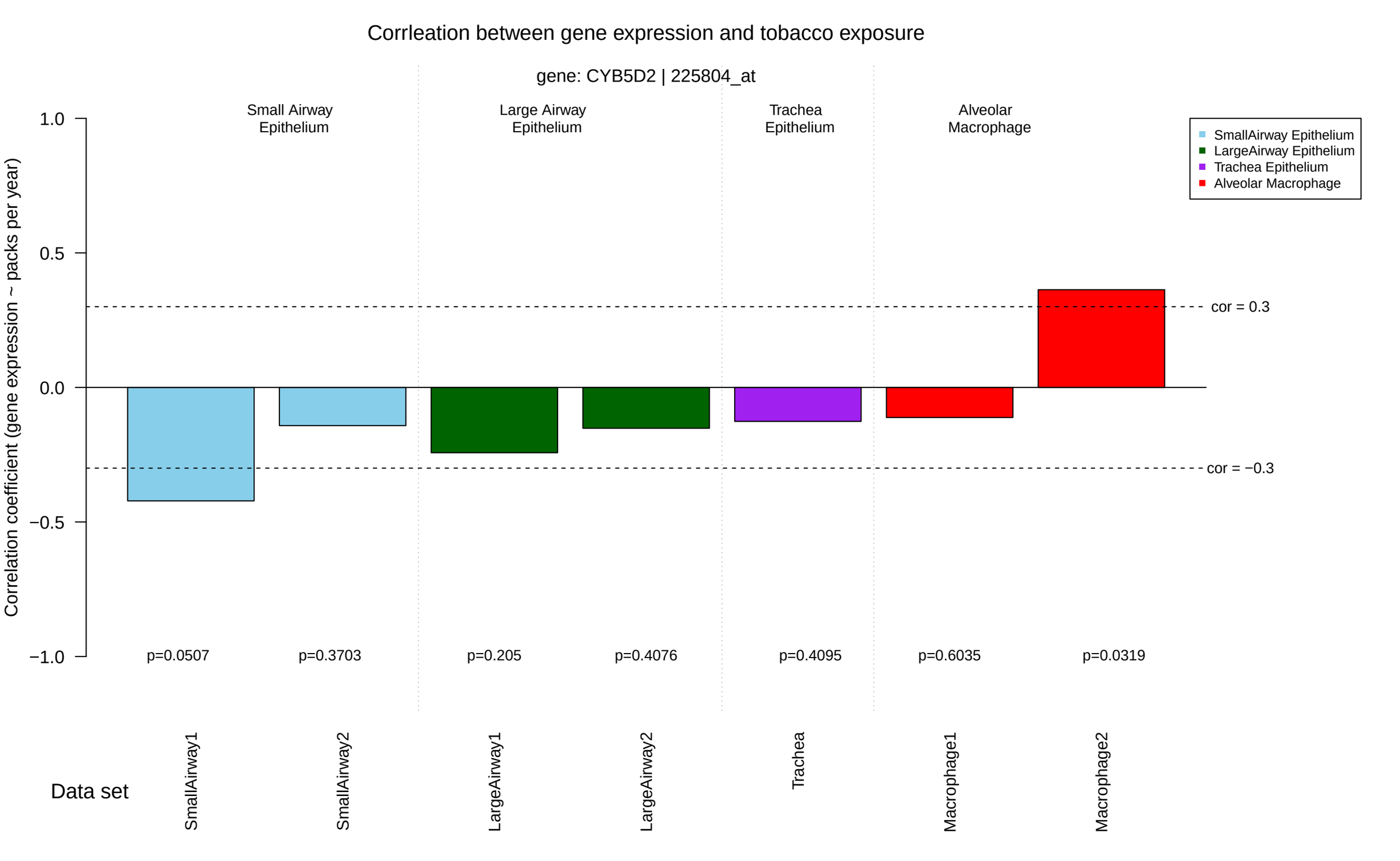

set: LargeAirway1 (p=0.0027)
CYB5D2 | 225804_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2766)
CYB5D2 | 225804_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.059)
CYB5D2 | 225804_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1228)
CYB5D2 | 225804_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2086)
CYB5D2 | 225804_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYB5D2 | 225804_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0507
p=0.3703
p=0.205
p=0.4076
p=0.4095
p=0.6035
p=0.0319
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2